

Finite-Cotype De-Poissonization and the UMD Case of Remark 12.10

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Status

This packet gives a candidate full solution, subject to human review, to the UMD general-local part of Remark 12.10 in Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. The source proves the quasi-left-continuous case for UMD spaces and says that the remaining issue for general local martingales is a discrete independent-increment comparison. We prove the finite independent-increment comparison in every Banach space of finite cotype. This is the form needed in the accessible-jump approximation in the source: the accessible part is first truncated to finitely many predictable jumps and then one lets the truncation parameter tend to infinity. Since every UMD space has finite cotype, this settles the UMD case of the source question.

This does not contradict the previous c_0 counterexample in the run: c_0 has no finite cotype and is not UMD.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source states that the general local-martingale case is open and reduces the missing issue to independent increments.

(12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation states the aggregate-domination formulation and notes that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Main Results

Theorem 1 (Finite-cotype discrete aggregate comparison). *Let X be a Banach space of finite cotype, and let $1 \leq p < \infty$. Let $(\xi_i)_{i=1}^n$ and $(\eta_j)_{j=1}^m$ be independent mean-zero X -valued random variables with finite p -moments. Suppose that*

$$\sum_i \mathbb{P}(\xi_i \in B) \leq \sum_j \mathbb{P}(\eta_j \in B) \quad (B \in \mathcal{B}(X \setminus \{0\})).$$

Then

$$\left(\mathbb{E} \left\| \sum_i \xi_i \right\|^p \right)^{1/p} \leq C_{p,X} \left(\mathbb{E} \left\| \sum_j \eta_j \right\|^p \right)^{1/p}.$$

Theorem 2 (UMD general local characteristic domination). *Let X be a UMD Banach space and $1 \leq p < \infty$. If N and M are X -valued local martingales and N is characteristically dominated by M in the sense of [1, Definition 12.9], then*

$$\mathbb{E} \sup_{t \geq 0} \|N_t\|^p \leq C_{p,X} \mathbb{E} \sup_{t \geq 0} \|M_t\|^p.$$

Relation to the c_0 Counterexample

There are two closely related questions in Remark 12.10. The broad discrete aggregate-domination question is phrased for an arbitrary Banach space X . That broad question is false: the companion packet

`1907.11588.discrete_aggregate_domination_c0_counterexample`

constructs independent symmetric c_0 -valued families with equal aggregate jump measures but with unbounded comparison constants. This is not a counterexample to the present theorem, because c_0 has no finite cotype and is not UMD.

The present packet proves the positive UMD/finite-cotype side. Thus the two packets should be read as complementary boundary information: arbitrary Banach spaces are too large, while finite cotype is enough for the discrete comparison proved here, and UMD spaces fall inside that positive class.

Idea of the Proof

The earlier conditional packet in this run showed that the symmetric aggregate-domination problem follows from symmetric de-Poissonization: compare the original independent Bernoulli family to the compound-Poisson sum with the same aggregate nonzero law. The new observation is that finite cotype gives exactly the missing upper de-Poissonization estimate.

The key input is a Maurey–Pisier contraction principle. If X has cotype $q < \infty$, then iid symmetric scalar coefficients θ_i satisfy

$$\left(\mathbb{E} \left\| \sum_i \theta_i x_i \right\|^p \right)^{1/p} \lesssim_{p,q,X} \int_0^\infty \mathbb{P}(|\theta_1| > t)^{1/\max(p,q)} dt \left(\mathbb{E} \left\| \sum_i \varepsilon_i x_i \right\|^p \right)^{1/p}.$$

For Bernoulli selectors of density a , the tail integral is $a^{1/\max(p,q)}$. Hence an a -thinning of a symmetric independent family costs a summable power of a . A compound-Poisson sum decomposes into layers $\{\text{Pois}(p_i) \geq k\}$, and the k -th layer is a $1/k!$ -thinning of the original family. Summing over k proves de-Poissonization.

A Finite-Cotype Thinning Lemma

We use the following standard Maurey–Pisier contraction principle in the explicit form stated by Ivanisvili–van Handel–Volberg [2, Theorem 4.1].

Theorem 3 (Maurey–Pisier contraction). *Let X have cotype $q < \infty$, let $\theta_1, \dots, \theta_n$ be iid symmetric scalar random variables, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher signs. Then for every $1 \leq p < \infty$ and every $x_1, \dots, x_n \in X$,*

$$\left(\mathbb{E} \left\| \sum_i \theta_i x_i \right\|^p \right)^{1/p} \leq L_{p,q,X} \int_0^\infty \mathbb{P}(|\theta_1| > t)^{1/\max(p,q)} dt \left(\mathbb{E} \left\| \sum_i \varepsilon_i x_i \right\|^p \right)^{1/p}.$$

Lemma 1 (Thinning in finite cotype). *Let X have cotype $q < \infty$, let $1 \leq p < \infty$, and set*

$$\alpha = \frac{1}{\max(p,q)}.$$

Let $Z_1, \dots, Z_n$ be independent symmetric X -valued random variables, and let $R_1, \dots, R_n$ be independent Bernoulli variables, independent of the Z_i , with $\mathbb{P}(R_i = 1) \leq a \leq 1$. Then

$$\left(\mathbb{E} \left\| \sum_i R_i Z_i \right\|^p \right)^{1/p} \leq K_{p,q,X} a^\alpha \left(\mathbb{E} \left\| \sum_i Z_i \right\|^p \right)^{1/p}.$$

Proof. First suppose $x_1, \dots, x_n \in X$ are deterministic and that R_i are iid Bernoulli a . Apply Theorem 3 to $\theta_i = R_i \varepsilon_i$. Since

$$\mathbb{P}(|\theta_1| > t) = \begin{cases} a, & 0 \leq t < 1, \\ 0, & t \geq 1, \end{cases}$$

the tail integral equals a^α . Thus

$$\left(\mathbb{E} \left\| \sum_i R_i \varepsilon_i x_i \right\|^p \right)^{1/p} \leq K_{p,q,X} a^\alpha \left(\mathbb{E} \left\| \sum_i \varepsilon_i x_i \right\|^p \right)^{1/p}.$$

If the Bernoulli parameters are $\rho_i \leq a$, realize a Bernoulli a signed selector as the smaller selector plus an independent centered remainder. First select R_i with parameter ρ_i and sign ε_i . If $\rho_i < 1$, then on the event $R_i = 0$ add an extra selector E_i with conditional probability $(a - \rho_i)/(1 - \rho_i)$, and give this extra selector a fresh independent sign δ_i . If $\rho_i = 1$, necessarily $a = 1$ and no extra selector is needed. The scalar

$$\Theta_i = R_i \varepsilon_i + (1 - R_i) E_i \delta_i$$

is symmetric and has the Bernoulli a signed-selector law. Moreover, conditionally on the smaller thinning $(R_i, \varepsilon_i)_i$, the extra sum has mean zero. Jensen's inequality for the convex function $x \mapsto \|x\|^p$ gives

$$\left(\mathbb{E} \left\| \sum_i R_i \varepsilon_i x_i \right\|^p \right)^{1/p} \leq \left(\mathbb{E} \left\| \sum_i \Theta_i x_i \right\|^p \right)^{1/p},$$

so the same bound holds for arbitrary $\rho_i \leq a$.

For random symmetric Z_i , condition on their values z_i and insert independent Rademacher signs. Because Z_i is symmetric, $(\varepsilon_i Z_i)_i$ has the same joint law as $(Z_i)_i$, and similarly after thinning. Integrating the deterministic estimate over the z_i 's proves the lemma. $\square$

Finite-Cotype De-Poissonization

For a finite independent symmetric family $Z_1, \dots, Z_n$, write

$$\mu_Z = \sum_i \mathcal{L}(Z_i)|_{X \setminus \{0\}}.$$

Let Π_{μ_Z} be the compound-Poisson sum with intensity μ_Z .

Proposition 1 (Symmetric de-Poissonization in finite cotype). *Let X have finite cotype and $1 \leq p < \infty$. Then*

$$(\mathbb{E} \|\Pi_{\mu_Z}\|^p)^{1/p} \leq D_{p,X} \left(\mathbb{E} \left\| \sum_i Z_i \right\|^p \right)^{1/p}$$

for every finite independent symmetric X -valued family Z_i .

Proof. Let $p_i = \mathbb{P}(Z_i \neq 0)$. If $p_i > 0$, write U_i for the conditional law of Z_i given $Z_i \neq 0$; this conditional law remains symmetric. Let N_i be independent Poisson variables with means p_i , and let $(U_{i,k})_{k \geq 1}$ be independent copies of U_i . Then

$$\Pi_{\mu_Z} = \sum_i \sum_{k=1}^{N_i} U_{i,k} = \sum_{k \geq 1} L_k, \quad L_k = \sum_i \mathbf{1}_{\{N_i \geq k\}} U_{i,k}.$$

By Minkowski's inequality in $L^p(\Omega; X)$,

$$\|\Pi_{\mu_Z}\|_{L^p} \leq \sum_{k \geq 1} \|L_k\|_{L^p}.$$

For $0 \leq s \leq 1$,

$$\mathbb{P}(\text{Pois}(s) \geq k) \leq \mathbb{E} \binom{\text{Pois}(s)}{k} = \frac{s^k}{k!} \leq \frac{s}{k!}.$$

Thus, if $p_i > 0$, the i -th summand in L_k is distributed as $R_{i,k} Z_i$ with thinning parameter

$$\rho_{i,k} = \frac{\mathbb{P}(\text{Pois}(p_i) \geq k)}{p_i} \leq \frac{1}{k!};$$

the terms with $p_i = 0$ are zero. Lemma 1 therefore gives

$$\|L_k\|_{L^p} \leq K_{p,X} (1/k!)^\alpha \left(\mathbb{E} \left\| \sum_i Z_i \right\|^p \right)^{1/p},$$

where $\alpha = 1/\max(p, q) > 0$ for some finite cotype q of X . Since $\sum_{k \geq 1} (1/k!)^\alpha < \infty$, summing over k proves the proposition. $\square$

Aggregate Domination

Lemma 2 (Lower Poissonization). *For every Banach space X , every $1 \leq p < \infty$, and every finite independent symmetric family Z_i ,*

$$\left(\mathbb{E} \left\| \sum_i Z_i \right\|^p \right)^{1/p} \leq 2 (\mathbb{E} \|\Pi_{\mu_Z}\|^p)^{1/p}.$$

Proof. This is the lower Poissonization lemma from the companion conditional packet, but we recall the proof. Write $Z_i = B_i U_i$, where B_i is Bernoulli $p_i = \mathbb{P}(Z_i \neq 0)$ and U_i has the symmetric conditional law of Z_i given $Z_i \neq 0$. Let R_i be iid Bernoulli variables with

$$c = 1 - e^{-1} > 1/2,$$

independent of everything else, and put $S = \sum_i B_i U_i$, $S_R = \sum_i R_i B_i U_i$. The complement $\sum_i (1 - R_i) B_i U_i$ is a $(1 - c)$ -thinning of the original family. Since $1 - c < c$, it can be obtained as a sub-thinning of a c -thinning; conditioning on the smaller thinning and adding the independent symmetric remainder back gives, by Jensen,

$$\left\| \sum_i (1 - R_i) B_i U_i \right\|_{L^p} \leq \|S_R\|_{L^p}.$$

The triangle inequality gives $\|S\|_{L^p} \leq 2\|S_R\|_{L^p}$.

Now let N_i be independent Poisson variables with means p_i . Since

$$\mathbb{P}(N_i \geq 1) = 1 - e^{-p_i} \geq (1 - e^{-1})p_i = cp_i, \quad 0 \leq p_i \leq 1,$$

the first Poisson layer $\sum_i \mathbf{1}_{\{N_i \geq 1\}} U_{i,1}$ contains, in the same Jensen/sub-thinning sense, a copy of S_R . Thus

$$\|S_R\|_{L^p} \leq \left\| \sum_i \mathbf{1}_{\{N_i \geq 1\}} U_{i,1} \right\|_{L^p}.$$

Finally, the full compound-Poisson sum is this first layer plus further conditionally mean-zero symmetric layers, so Jensen gives that the first layer is controlled by Π_{μ_Z} . Combining the estimates proves the claim. $\square$

Lemma 3 (Poisson monotonicity). *Let $\mu \leq \nu$ be finite symmetric Borel measures on $X \setminus \{0\}$. Then*

$$\|\Pi_\mu\|_{L^p} \leq \|\Pi_\nu\|_{L^p}.$$

Proof. By Poisson superposition, Π_ν has the same law as $\Pi_\mu + \Pi_{\nu-\mu}$, with the two terms independent. Since $\nu - \mu$ is symmetric, $\Pi_{\nu-\mu}$ is mean zero after truncation. Conditioning on Π_μ and applying Jensen's inequality gives the truncated claim, and then monotone convergence removes the truncation. $\square$

Proposition 2 (Symmetric aggregate comparison in finite cotype). *Theorem 1 holds when all ξ_i, η_j are symmetric.*

Proof. Let $\mu_\xi = \sum_i \mathcal{L}(\xi_i)|_{X \setminus \{0\}}$ and $\mu_\eta = \sum_j \mathcal{L}(\eta_j)|_{X \setminus \{0\}}$. By assumption $\mu_\xi \leq \mu_\eta$. Lemma 2, Lemma 3, and Proposition 1 give

$$\left\| \sum_i \xi_i \right\|_{L^p} \leq 2 \|\Pi_{\mu_\xi}\|_{L^p} \leq 2 \|\Pi_{\mu_\eta}\|_{L^p} \leq 2D_{p,X} \left\| \sum_j \eta_j \right\|_{L^p}.$$

$\square$

Proof of Theorem 1. Let $\varepsilon_i, \varepsilon'_j$ be independent Rademacher signs, independent of all variables. Symmetrizing the laws preserves aggregate domination:

$$\sum_i \mathcal{L}(\varepsilon_i \xi_i)|_{X \setminus \{0\}} \leq \sum_j \mathcal{L}(\varepsilon'_j \eta_j)|_{X \setminus \{0\}}.$$

By the standard symmetrization inequalities for independent mean-zero Banach-valued variables,

$$\left\| \sum_i \xi_i \right\|_{L^p} \leq 2 \left\| \sum_i \varepsilon_i \xi_i \right\|_{L^p}, \quad \left\| \sum_j \eta_j \right\|_{L^p} \leq 2 \left\| \sum_j \varepsilon'_j \eta_j \right\|_{L^p}.$$

Apply Proposition 2 to the symmetric families $(\varepsilon_i \xi_i)$ and $(\varepsilon'_j \eta_j)$, and combine the three estimates. $\square$

General Local Martingales

Proof of Theorem 2. Yaroslavtsev’s quasi-left-continuous characteristic-domination theorem [1, Theorem 12.6] handles the continuous and quasi-left-continuous parts in UMD spaces. In Remark 12.10, using the detangling theorem for martingales with accessible jumps and the accessible approximation proposition, the source identifies the remaining issue for general local martingales as the discrete independent-increment aggregate-domination problem stated there. The finite form proved above is the one used at the approximation level: the source’s accessible-jump approximation truncates a purely discontinuous accessible martingale to finitely many predictable jumps M^m , proves the desired finite discrete estimate for those jumps, and then lets $m \rightarrow \infty$ in L^p .

Every UMD Banach space has finite cotype. Therefore Theorem 1 supplies the missing accessible-jump comparison in X . Substituting this comparison into the source reduction gives the maximal L^p estimate for general local martingales. $\square$

Verification and Novelty Notes

The cheap run indexes were searched for 1907.11588, Remark 12.10, aggregate domination, UMD, de-Poissonization, Rademacher, finite cotype, and Maurey–Pisier. The run already contained the c_0 counterexample, Hilbert all- p packets, $L^q(S)$ packets, and the conditional de-Poissonization reduction. No prior packet contained the finite-cotype thinning argument above.

The decisive external input is the Maurey–Pisier contraction principle as quoted in [2, Theorem 4.1]; the parsed source lines 2003.06345/source.tex:803--858 state the exact theorem used here. The fact that UMD spaces have finite cotype is standard in Banach-space probability, and is also recalled in the local corpus in Hytönen–McIntosh–Portal, arXiv:0703012, lines 367–372.

The verification pass checked the following source joins. The exact Maurey–Pisier estimate used in Lemma 1 is [2, Theorem 4.1]. Yaroslavtsev’s Remark 12.10 states the open independent-increment aggregate-domination problem and explicitly says that the symmetric case was not known; the surrounding proof of the accessible-jump estimates uses finite predictable-jump approximants and the approximation proposition for accessible martingales.

Human review should focus on Lemma 1, especially the passage from iid Bernoulli selectors to non-identical sub-thinnings, and on the final appeal to the accessible-jump approximation in Yaroslavtsev’s reduction.

References

- [1] I. S. Yaroslavtsev. *Local characteristics and tangency of vector-valued martingales*. arXiv:1907.11588, 2019.
- [2] P. Ivanisvili, R. van Handel, and A. Volberg. *Rademacher Type and Enflo Type Coincide*. arXiv:2003.06345, 2020.
- [3] M. Ledoux and M. Talagrand. *Probability in Banach Spaces*. Springer, 1991.
- [4] T. Hytönen, A. McIntosh, and P. Portal. *Kato’s square root problem in Banach spaces*. arXiv:0703012, 2007.